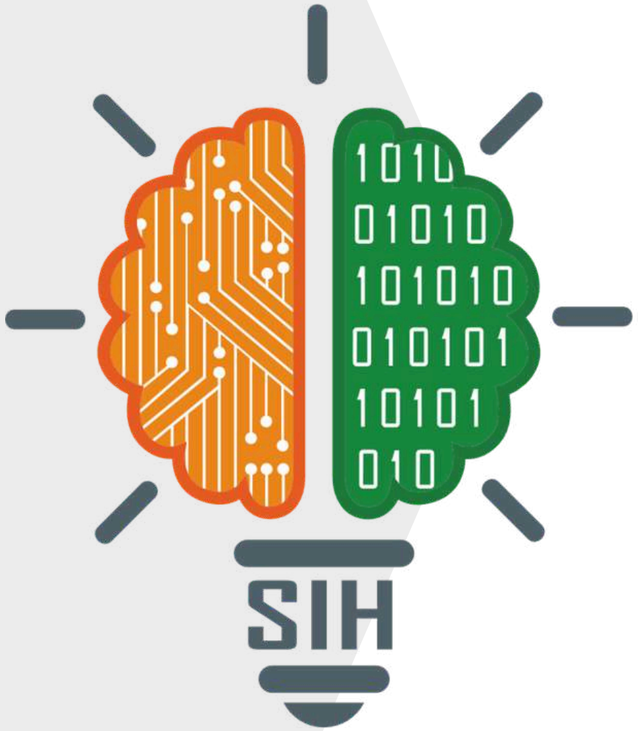
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Flat to 3D

Single-View Height Estimation and 3D Flythrough

- **Problem Statement ID – SIH26175**
- **Problem Statement Title- DepthWizard - Single-View Height Estimation and 3D Flythrough**
- **Theme- Disaster Management**
- **PS Category- Software**
- **Team ID- 95**
- **Team Name : Ghazal**



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From Flat to 3D
Calibrating AI Depth Models with Real-World Geographic Constraints

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PROBLEM EXISTING

- **RGB images lack direct height information,** making real-world elevation difficult to recover from a single image.
- Monocular AI models estimate **relative depth**, but their predictions can lack reliable real-world scale.
- Conventional accurate methods such as **LiDAR and multi-view photogrammetry** require additional sensors, imagery, or higher acquisition cost.

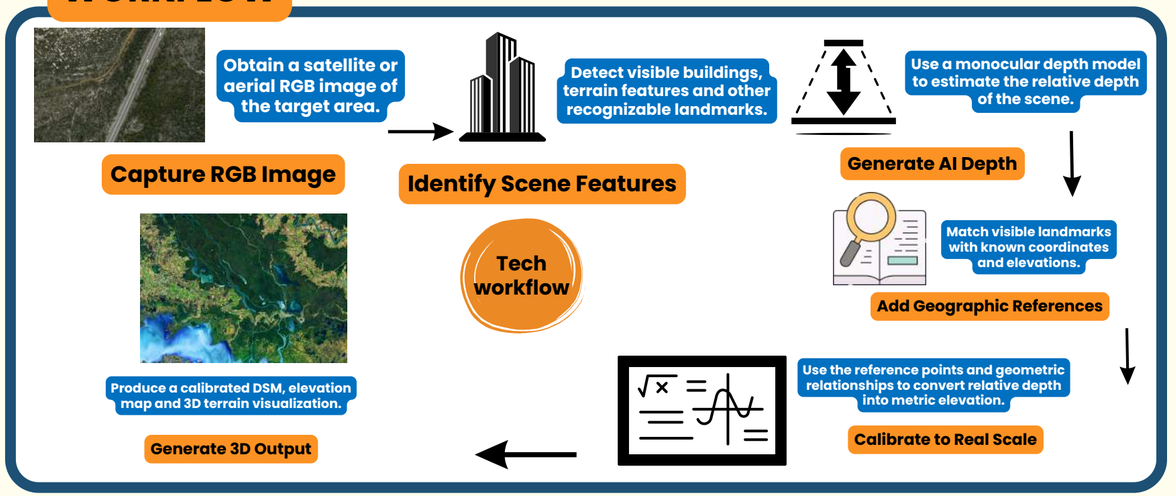
PROPOSED SOLUTION

- Use a **monocular AI depth model** to generate a relative depth map from a single RGB image.
- Match visible scene features with a **geographic reference database** containing known coordinates and elevations.
- Apply **metric calibration** using reference elevations and geometric constraints to convert relative depth into real-world scale.
- Generate the calibrated output as a **Digital Surface Model (DSM) and 3D elevation visualization.**

UVP (UNIQUE VALUE PROPOSITION)

- **Single-Image Metric Depth**: Converts monocular relative depth into real-world elevation.
- **Geographic Grounding**: Uses known reference points to anchor AI predictions to measurable coordinates and heights.
- **Low-Cost & Scalable**: Reduces dependence on LiDAR and complex multi-view sensing while retaining geographic verifiability.

WORKFLOW



ARCHITECTURE

- **Input & Depth Estimation:** The RGB image is processed by a monocular AI model to generate a r**elative depth map** representing the scene's depth structure.
- **Reference Matching**: Visible landmarks are matched with **known geographic coordinates and real-world elevations** to provide reliable reference points.
- **Metric Calibration**: The relative depth is calibrated against the reference measurements and **geometric constraints** to obtain real-world metric elevation.
- **Output Generation:** The calibrated elevation data is transformed into a **DSM, 3D mesh, and elevation map** for visualization and analysis.

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Technical Approach

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TECHSTACK

MAGE INPUT & PROCESSING

Python + OpenCV

Input: single RGB satellite/aerial image
Image resizing and preprocessing
Extraction of image coordinates/control points

MONOCULAR DEPTH ESTIMATION

PyTorch + Metric3D

AI analyses the single RGB image
Generates a dense depth map
Provides the relative/metric depth information required for reconstruction

REAL-WORLD REFERENCE

DEM + GPS/Geographic Reference Points

Rasterio + GDAL + PyProj
Obtain known elevation values from a DEM
Match identifiable landmarks/control points with geographic coordinates

METRIC CALIBRATION & GEOMETRY

NumPy + SciPy + Camera Geometry

Performs the metric calibration, estimating the relationship between AI-predicted depth and real-world measurements through mathematical optimization.

3D RECONSTRUCTION

Open3D

Converts the calibrated depth/elevation information into a 3D point cloud, surface/DSM, and visualizable terrain model.

IMAGE RENDERING

Unity

Rendering the images with 3D caliberation and dimensions to estimate flythrough

PROTOTYPE LINK [Click here]

Technical Worklow



RGB Image Preprocessing

Input image → resize/normalize → prepare image for depth inference.

Monocular Depth Estimation

RGB image → depth model → relative depth map

Feature & Landmark Matching

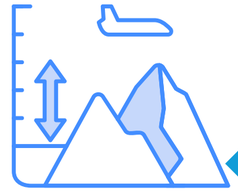
Detect image features → match with geographic reference database → obtain known coordinates + elevations

how it goes about



3D Reconstruction & Visualization

Depth-to-DSM Apply calibrated scale → transform depth values into real-world surface elevations



Depth-to-DSM Conversion

Relative depth + reference elevations + spatial/geometric constraints → estimate metric scale and elevation



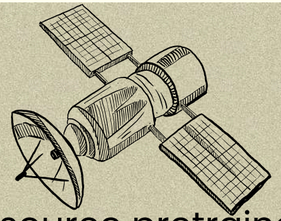
Geometric Calibration

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Feasibility and Viability

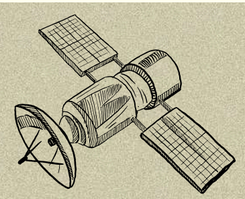
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Feasibility



- **Existing Technology:** Uses open-source pretrained depth models (MiDaS, Depth Anything), so no model training is needed.
- **Data Readiness:** Integrates with public elevation sources (SRTM, Copernicus DEM) for GCPs, and camera intrinsics from image EXIF data.
- **Practical Build**: Calibration is a simple two-parameter fit that runs on a single image in Python.

Viability



- **Cost Efficiency:** Replaces LiDAR surveys and stereo capture with one photograph and a few known elevations.
- **Better Planning:** Quick building-height and terrain estimates support urban planning and flood or disaster assessment.
- **Measurable Trust:** Accuracy is verified with RMSE against held-out check points, so results can be validated.

Data & Technical Challenges

- **Scale Ambiguity:** Raw network depth has no physical units and needs GCPs to become metric.
- **GCP Quality:** Wrong or poorly spread reference points skew the scale, so we use RANSAC or a Huber loss.
- **Camera Intrinsics:** Wrong focal length or lens distortion distorts the X and Y coordinates.

Operational Challenges

- **GCP Availability:** Reliable elevation references may be scarce in remote areas.
- **Viewpoint Limits:** Oblique images need camera pose, and a single view cannot see occluded surfaces.
- **Difficult Surfaces:** Water, glass and sky give unreliable depth, so they are masked or interpolated.

AI & Machine Learning Challenges

- **Non-Linear Error:** One global affine fit assumes linear network error, so we use many GCPs or a piecewise calibration.
- **Domain Shift:** Models can perform worse on scenes unlike their training data, such as aerial or unusual terrain.
- **Depth vs Inverse Depth:** Some models (like MiDaS) output inverse depth and need conversion before fitting.

User Interface Challenges

- **Uncertainty Display:** Users need to see the error range (RMSE) so results are not over-trusted.
- **GCP Selection:** Non-experts need simple tools to pick and match landmarks.
- **3D Visualization:** Large point clouds and DSM grids must render smoothly in the browser.

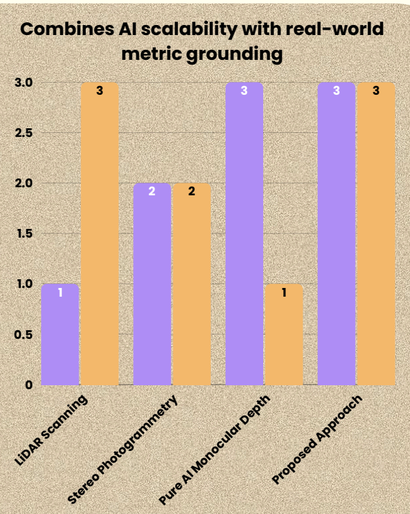
In-Depth Estimation and Caliberation report: Click here

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Impact and Benefits

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- **Cost & Deployment:** Uses a single RGB satellite/aerial image with existing geographic references, reducing dependence on LiDAR or multi-view imaging.
- **Metric Grounding:** Converts AI-generated relative depth into real-world elevation using known landmarks, GPS coordinates, baseline elevations and geometric constraints.
- **Practical 3D Intelligence:** Produces metric height maps and interactive 3D terrain models for elevation querying, surface analysis and building inspection.



Feature	Existing	Proposed
Single RGB image	Limited	☐
Real-world metric elevation	Limited	☐
Geographic reference points	—	☐
Geometric calibration	—	☐
3D terrain / height map	Partial	☐
No LiDAR / multi-view requirement	—	☐

How we work?

SCENARIO 1: Standard Image + Known References

For: A clear RGB satellite/aerial image with identifiable geographic landmarks.
Process: RGB Image → Monocular Depth → Reference Matching → Metric Calibration → 3D Elevation.
Output :Metric-grounded height information and 3D terrain.

SCENARIO 2: Complex Elevation / Terrain

For: Scenes where relative depth alone cannot provide reliable real-world scale.
Process: Landmark elevations + GPS references → geometric constraints → calibrated depth.
Output: Refined metric elevations using real-world spatial relationships.

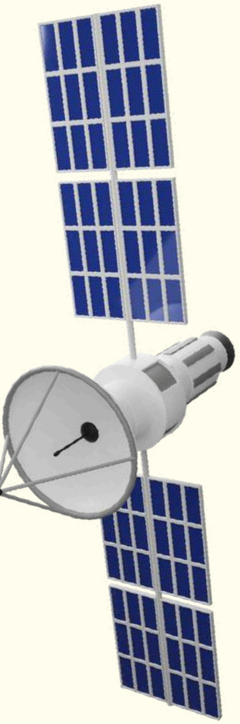
SCENARIO 3: Limited Geographic References

For: Images where reliable reference coverage is insufficient for full metric calibration.
Process: Monocular model still generates relative depth → metric scale applied where valid references are available.
Output: Useful depth visualization with metric grounding dependent on reference availability.

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Research and References

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- R. Ranftl, K. Lasinger, D. Hafner, K. Schindler, V. Koltun, "Towards Robust Monocular Depth Estimation: Mixing Datasets for Zero-shot Cross-dataset Transfer," IEEE TPAMI, 2022. arXiv:1907.01341. This is the MiDaS paper. It proposes a training objective that is invariant to changes in depth range and scale, which is why the output is relative and needs GCP calibration. arxiv
- R. Ranftl, A. Bochkovskiy, V. Koltun, "Vision Transformers for Dense Prediction," ICCV, 2021, pp. 12179-12188. DOI: 10.1109/ICCV48922.2021.01196. This is the DPT architecture used in later MiDaS versions and in Depth Anything.
- L. Yang, B. Kang, Z. Huang, X. Xu, J. Feng, H. Zhao, "Depth Anything: Unleashing the Power of Large-Scale Unlabeled Data," CVPR, 2024, pp. 10371-10381. DOI: 10.1109/CVPR52733.2024.00987. Its supplement states that in zero-shot evaluation the prediction's scale and shift are aligned to the ground truth, the same alignment step as our $Z = s \cdot d + t$. thecvf
- L. Yang, B. Kang, Z. Huang, Z. Zhao, X. Xu, J. Feng, H. Zhao, "Depth Anything V2," NeurIPS, 2024. arXiv:2406.09414. DOI: 10.52202/079017-0688.

3. METHODOLOGY

An AI model analyzes visual cues to produce a relative depth map. This establishes scene shape ("A is taller than B") but lacks real-world metric height.

PROPOSED ARCHITECTURE & CALIBRATION

- **Reference Point Matching:** The system cross-references the image footprint with the landmark database to extract true surface elevations.
- **Metric Calibration:** Correlates AI relative depth with actual metric elevations using mathematical mapping (Pythagoras) (e.g., $Elevation = a \times Depth + b$).
- **Geometric Constraints:** Enforces structural accuracy using line-of-sight geometry and baseline low-resolution DEM data.

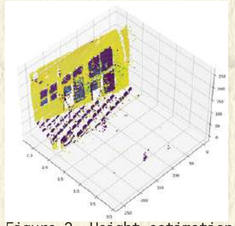


Figure 3. Height estimation

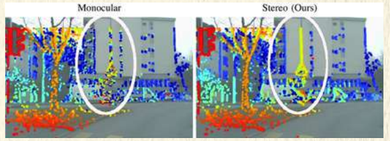


Figure 4. Monocular comparison

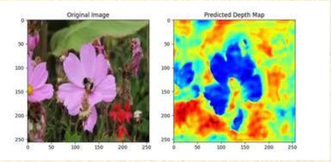
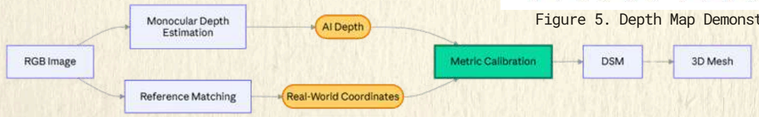


Figure 5. Depth Map Demonstration



This system bridges visual AI and rigorous geospatial science. Rather than relying on AI to "guess" heights, it extracts relative 3D structure and uses a network of geographic reference points for real-world metric calibration. Furthermore, the methodology integrates Pythagorean geometric constraints—utilizing line-of-sight slant and horizontal distances—to rigorously refine and verify the vertical elevation estimates.

Github Link

Click here